ICLR 2021

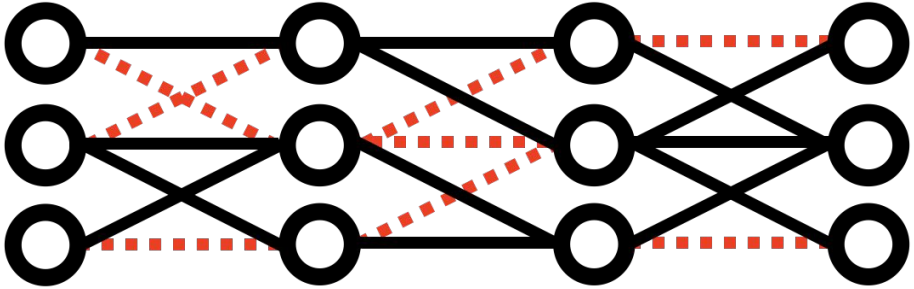
Layer-Adaptive Sparsity for the Magnitude-based Pruning

Jaeho Lee[♦] **Sejun Park**[♠] **Sangwoo Mo**[♦] **Sungsoo Ahn**[♣] **Jinwoo Shin**[♦]

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Background: MP and Layerwise Sparsity

Pruning. An art of removing unnecessary connections from big nets.



(usually for storage / inference efficiency)

Q. How do we know (before removing) if the target connection is not needed?

A. We don't; but we know that some scores work better than random.

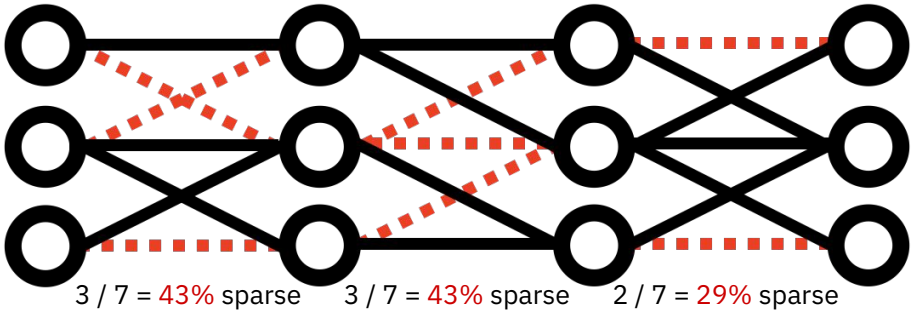
Proxy for saliency

Background: MP and Layerwise Sparsity

MP. Many scores have been proposed,
but people found that magnitude-based pruning (MP) works unreasonably well!

(See Gale et al. 2019)

But. MP works well only when we have the right Layerwise Sparsity!



Using (100%, 0%, 0%) sparsity won't work well, right?

> To understand the limits of the magnitude pruning heuristic, we modify our ResNet-50 training setup to leave the first convolutional layer fully dense, and only prune the final fully-connected layer to 80% sparsity. This heuristic is reasonable for ResNet-50, as the first layer makes up a small

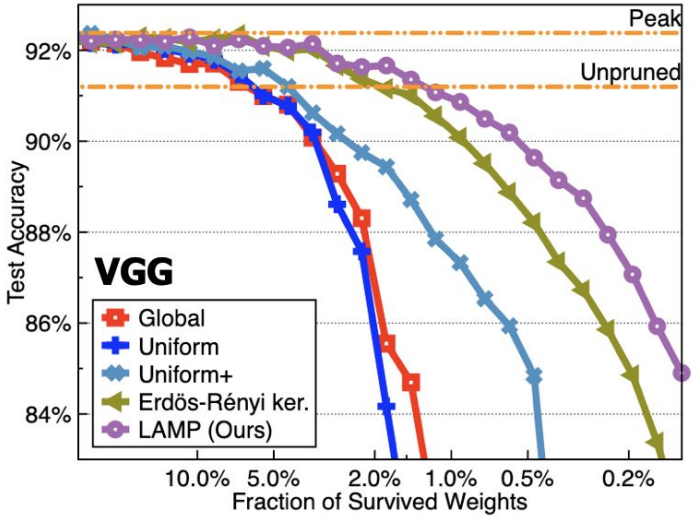
Gale et al. "The state of sparsity in deep neural networks," 2019.

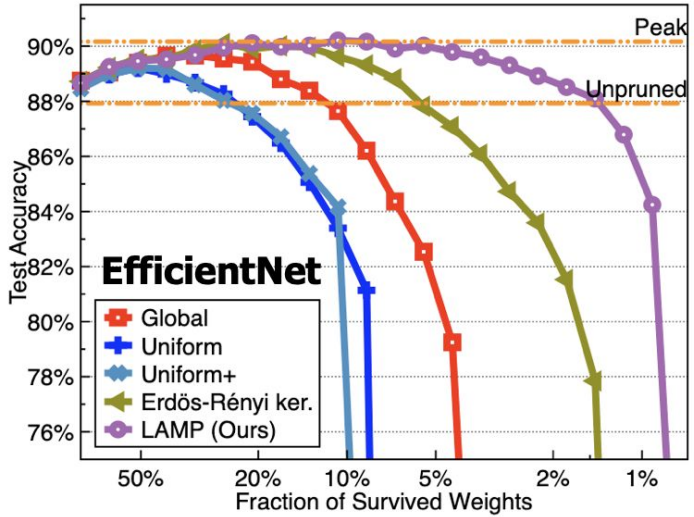
Question: What is the right layerwise sparsity for MP?

Contribution. In this work, we

(1) Compare existing layerwise sparsity determination schemes for MP.

(2) Propose a method that works well: **LAMP** Layer-adaptive MP





(Yes, we have more experiments in the paper)

Methods. LAMP in a nutshell.

Idea. We build on the observation that "Layerwise MP minimizes output distortion."
In particular, layerwise MP minimizes the Frobenius distortion

$$\|A - \widetilde{A}\|_F^2 \approx \|A - \widetilde{A}\|_2^2 = \sup_{\|x\| \leq 1} \|Ax - \widetilde{A}x\|_2^2$$

MP minimizes this

Output distortion from the layer, w.r.t. Worst-case output.

(A denotes the unpruned layer, Ã denotes the pruned layer).

LAMP. Designed to minimize the model-level distortion.

$$\sup_{\|x\| \leq 1} \|f(x; A_{1:d}) - f(x; \widetilde{A}_{1:d})\|_2^2$$

where

$$f(x; A_{1:d}) = A_d \circ \mathrm{ReLU} \circ A_{d-1} \circ \mathrm{ReLU} \circ \cdots \circ A_1 x$$

Methods. LAMP in a nutshell.

LAMP. LAMP applies a global threshold on the LAMP score.

For each A, assume that A is ordered: exists an index map such that $|A[u]| \leq |A[v]|$ holds whenever $u < v$. Then,

$$\text{score}(u; W) = \frac{(A[u])^2}{\sum_{v \geq u} (A[v])^2}$$

In other words, LAMP score is the mag^2 of connection u,
normalized by the mag^2 of same-layer connections that survive when pruning u.

Note. This is equivalent to doing MP layerwise, but an auto-tuned sparsity level.

Methods. LAMP in a nutshell.

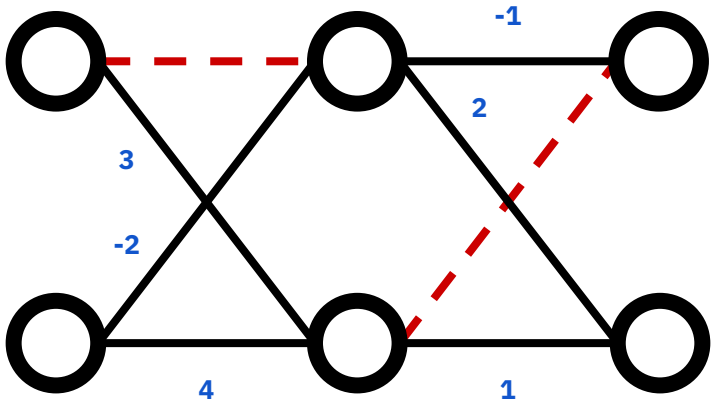
Roughly. In other words, pruning with LAMP is equivalent to

(1) Compute mag^2

(2) Normalize by $(\sum mag^2)$

(3) Prune a single connection from a model.

(4) Go to (1).



Methods. LAMP in a nutshell.

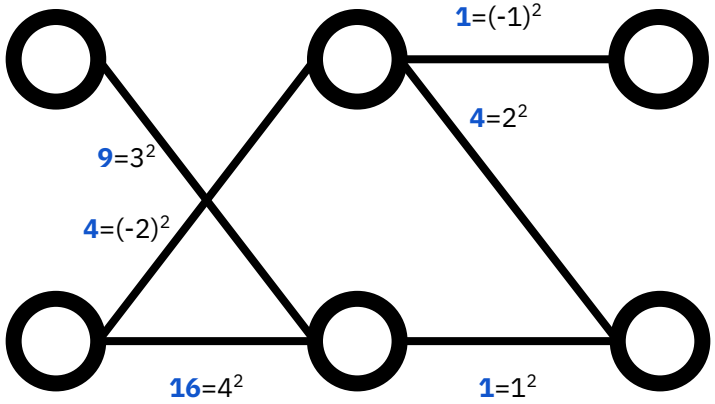
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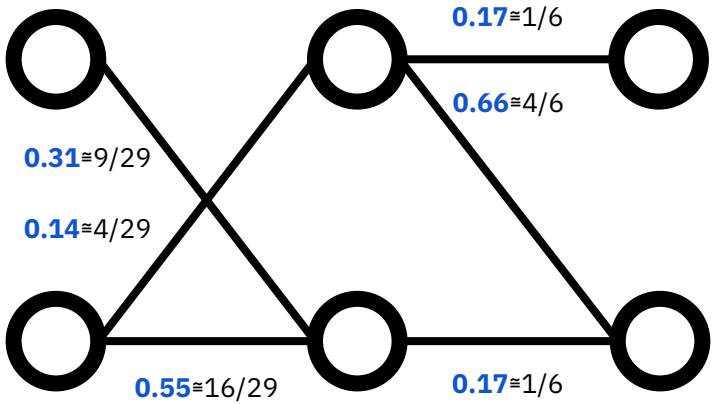
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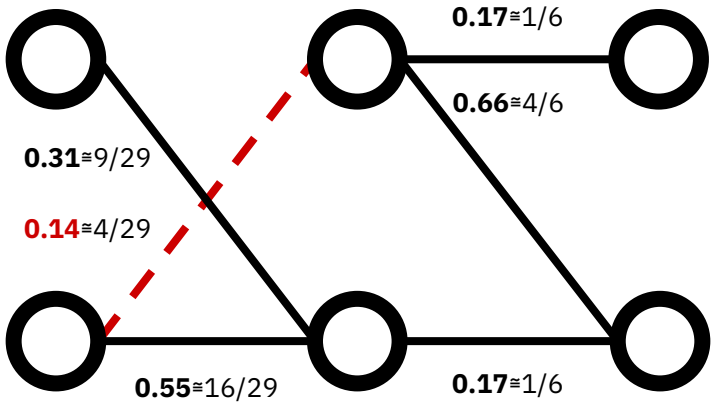
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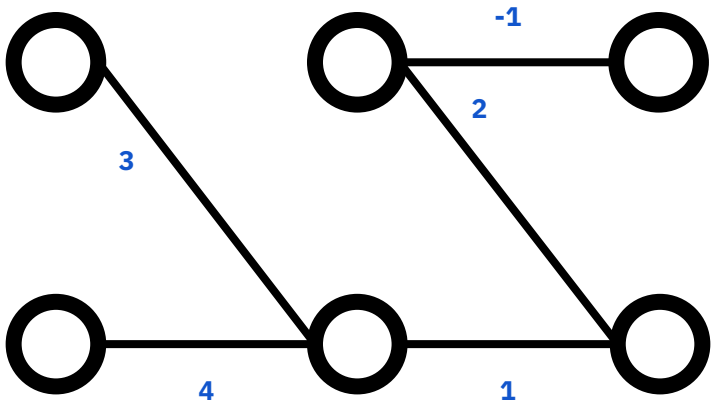
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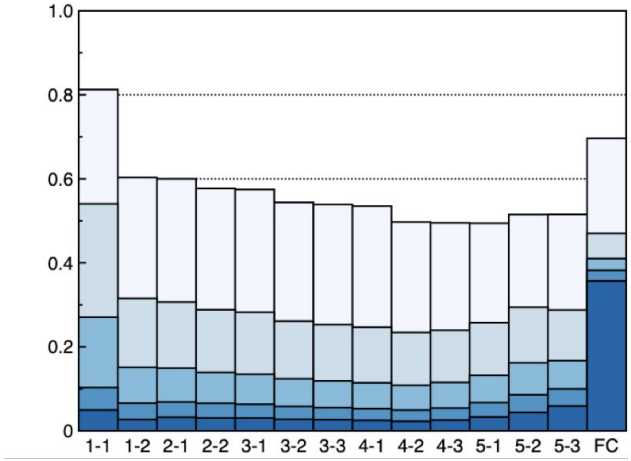
(4) Go to (1).



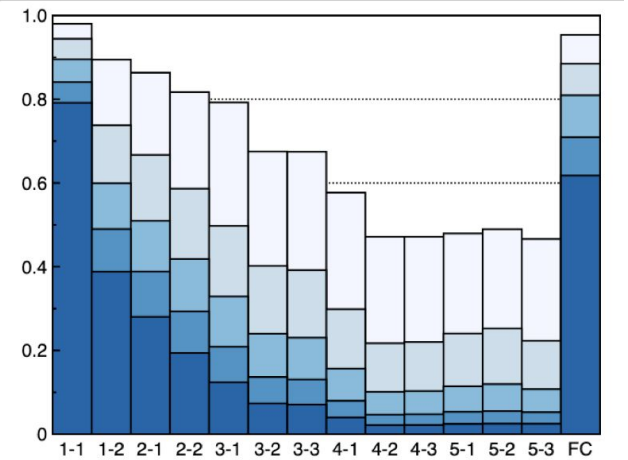
Note. LAMP computes everything in one step, using tensor operations.

Observation. What does LAMP discover?

Heuristic! LAMP not only performs well, but also recovers hand-crafted heuristics.



Survival Rate of VGG-16 + Global



Survival Rate of VGG-16 + LAMP

To understand the limits of the magnitude pruning heuristic, we modify our ResNet-50 training setup to leave the first convolutional layer fully dense, and only prune the final fully-connected layer to 80% sparsity. This heuristic is reasonable for ResNet-50, as the first layer makes up a small

Gale et al. "The state of sparsity in deep neural networks," 2019.

Poster. What to expect

Come to the Poster **#2494** at **Session 3** (PDT: May 3rd, 5pm-7pm)!

Chat with us about...

- **Baselines.** Some of them are cleverly designed, too.
- **Experiments.** LAMP also works well with rewinding.
- **Other properties.** LAMP is invariant to weight-rescaling.

Contact us via

- **Code repo.** https://github.com/jaeho-lee/layer-adaptive-sparsity
- **Mail.** jaeho-lee@kaist.ac.kr